

Poker Face Continuation

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Description

This project extends my first-semester group project, Poker Face, by significantly expanding both the technical depth of the AI and the complexity of the game environment. The goal of this project is to develop and evaluate multiple AI poker agents trained using various deep learning approaches, and to scale the game from a two-player setting to a full six-player Texas Hold'em table.

Specifically, I will:

- Explore alternative deep learning and reinforcement learning algorithms to train new poker agents.
- Implement a six-player Texas Hold 'em environment that supports any combination of human and AI players, from zero to six.
- Compare the performance and behavior of different AI models in multi-player scenarios, analyzing strengths, weaknesses, and emergent strategies.

The project will continue to use Python as the primary language, with PyTorch for model training and Pygame for the user interface. The final result will be an interactive poker application that allows users to play against multiple AI opponents with distinct strategies and varying skill levels.

Justification

This project is well-suited for a senior-level computer science capstone and aligns strongly with my personal interests in developing AI behavior in a game setting. Furthermore, Poker is a familiar yet strategically rich domain, and multi-agent AI systems are a central topic in modern artificial intelligence research. The project combines theory, implementation, and interactive demonstration, making it both technically meaningful and easy to appreciate.

It also gives me a clear opportunity to extend what I have already learned from testing and development in the first semester of this project. While the first-semester project focused on a two-player poker agent, this project introduces the significantly harder problem of multi-agent poker, where strategy must account for multiple opponents with differing behaviors. Training and comparing multiple deep learning approaches also pushes beyond standard classroom implementations and encourages deeper exploration of model design and evaluation.

The technical difficulty of this project is ambitious for a single semester and may require some trimming down. Expanding from two players to six players increases the complexity of every agent participating and has its own set of difficulties to set up, while training new models introduces nontrivial engineering and experimental challenges. However, because I am building on an existing codebase rather than starting from scratch, I believe I will be able to achieve my goal, even if it ends up in a slightly reduced final state to better fit the time constraints.

I already have experience with Python, PyTorch, and Pygame, as well as prior experience developing AI agents and game systems for this project. This minimizes startup time and allows me to focus on setting up a multiplayer network, higher-level design, and experimentation.